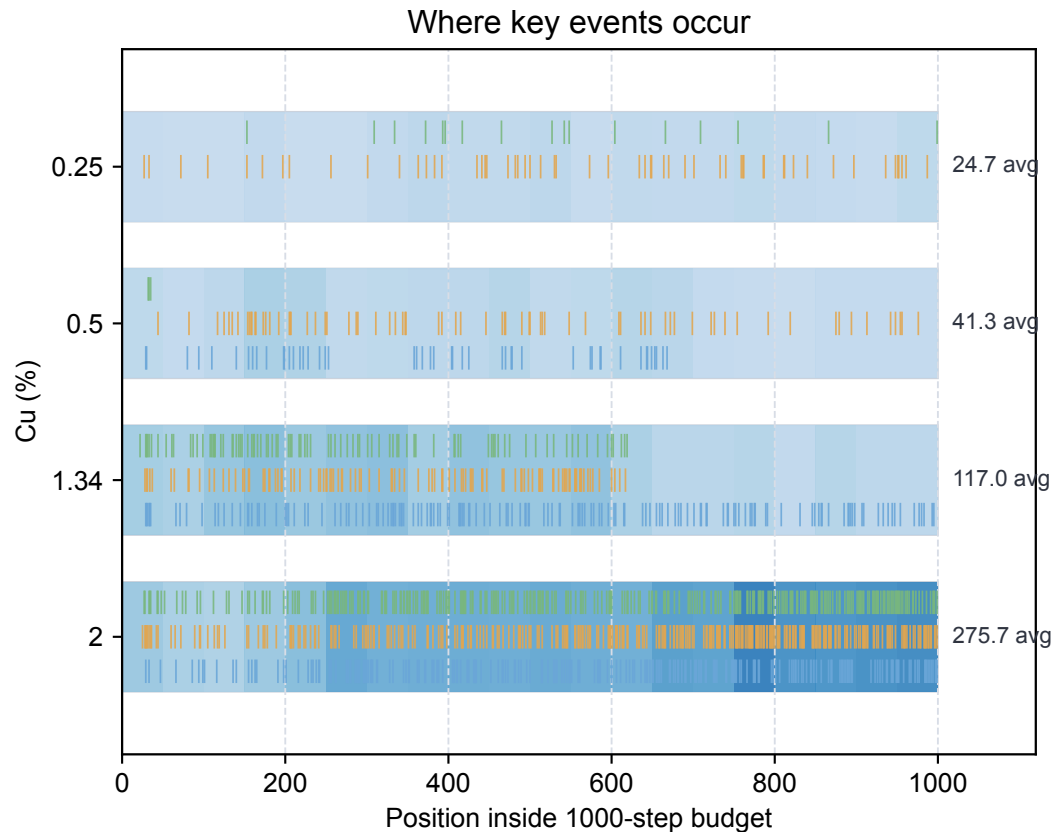
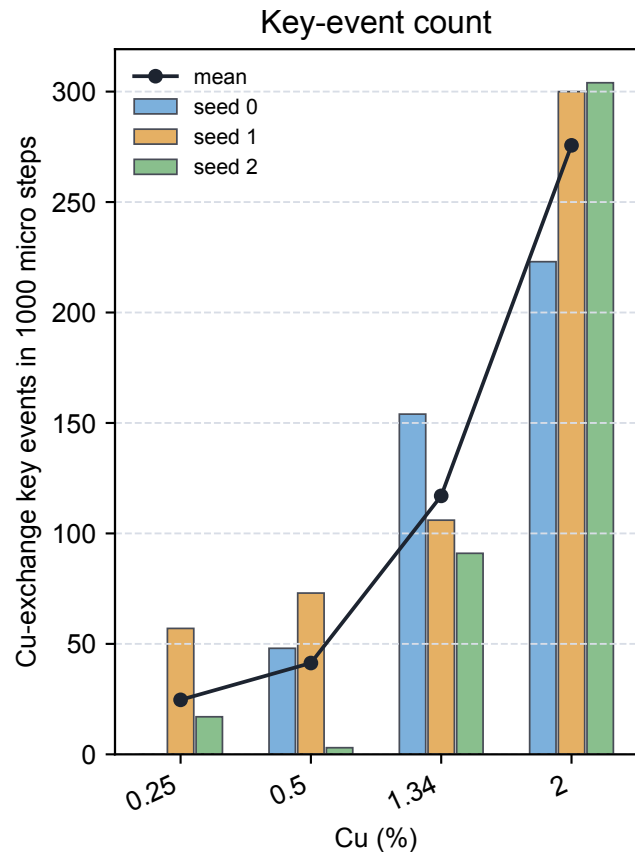


# 1000-step budget: Cu-density dependence of key structural exchanges



Diagnostic proxy: key evolution = Cu-vacancy exchange event. Colored ticks show seed-level randomness; blue bar intensity shows binned mean count.